Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2021 – Paper 2 – Solutions**

June 2021 – Question 1 – Solution

(a) (i) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be any two propositions. Complete the truth table below. [4]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{p}$	$\sim\boldsymbol{q}$	$\boldsymbol{p} \vee \boldsymbol{q}$	$\sim(\boldsymbol{p} \vee \boldsymbol{q})$	$\sim\boldsymbol{p} \vee \sim\boldsymbol{q}$
T	T	F	F	T	F	F
T	F	F	T	T	F	T
F	T	T	F	T	F	T
F	F	T	T	F	T	T

(ii) Hence, state whether the statements $\sim(\boldsymbol{p} \vee \boldsymbol{q})$ and $\sim\boldsymbol{p} \vee \sim\boldsymbol{q}$ are logically equivalent. Justify your response. [2]

The statements $\sim(p \vee q)$ and $\sim p \vee \sim q$ are not logically equivalent because their truth values are not the same.

(b) (i) Write the converse of the statement "$\boldsymbol{n}$ **is an integer** $\rightarrow$ $\boldsymbol{n}^2$ **is an integer**". [1]

Statement: "n is an integer $\rightarrow$ n^2 is an integer"

Converse: "n^2 is an integer $\rightarrow$ n is an integer"

(ii) For two real numbers x and y, the operation $*$ is given by $x * y = x^2 + y^2$. Prove that $*$ is closed in $\mathbb{R}$. [3]

For $x \in \mathbb{R},\ x^2 \in \mathbb{R}$. For $y \in \mathbb{R},\ y^2 \in \mathbb{R}$.

Also, addition is closed in $\mathbb{R}$.

Therefore, for any $x, y \in \mathbb{R}$,

$x * y = x^2 + y^2 \in \mathbb{R}$.

Hence, $*$ is closed in $\mathbb{R}$.

Q.E.D.

(iii) Determine whether the operation $*$ is commutative. [2]

$$x * y = x^2 + y^2$$
$$= y^2 + x^2$$
$$= y * x$$

Since $x * y = y * x$, then the operation $*$ is commutative.

(c) The function $f(x) = ax^3 + 3x^2 - b$ is divisible by $2x - 1$ and has a remainder of -5 when divided by $x + 2$.

Calculate the values of a and b. [7]

$f(x) = ax^3 + 3x^2 - b$

Since $f(x)$ is divisible by $(2x - 1)$, then by the Factor Theorem, $f\left(\frac{1}{2}\right) = 0$.

Well,

$f\left(\frac{1}{2}\right) = a\left(\frac{1}{2}\right)^3 + 3\left(\frac{1}{2}\right)^2 - b$

$0 = \frac{1}{8}a + \frac{3}{4} - b$

$b = \frac{1}{8}a + \frac{3}{4}$ $\rightarrow$ Equation 1

Since $f(x)$ has a remainder of -5 when divided by $(x+2)$, then by the Remainder Theorem, $f(-2) = -5$.

Well,

$f(-2) = a(-2)^3 + 3(-2)^2 - b$

$-5 = -8a + 3(4) - b$

$-5 = -8a + 12 - b$

$b = -8a + 17$ $\rightarrow$ Equation 2

Equating Equation 1 and Equation 2 gives:

$\frac{1}{8}a + \frac{3}{4} = -8a + 17$

$8a + \frac{1}{8}a = 17 - \frac{3}{4}$

$\frac{65}{8}a = \frac{65}{4}$

$\frac{1}{8}a = \frac{1}{4}$

$a = \frac{1}{4} \times 8$

$a = 2$

Substituting $a = 2$ into Equation 2 gives:

$b = -8(2) + 17$

$= -16 + 17$

$= 1$

$\therefore a = 2$ and $b = 1$

(d) Solve the logarithmic equation $log_2 x + log_4 x + log_{16} x = 7$. [6]

$log_2 x + log_4 x + log_{16} x = 7$

$log_2 x + \frac{log_2 x}{log_2 4} + \frac{log_2 x}{log_2 16} = 7$

$log_2 x + \frac{log_2 x}{2} + \frac{log_2 x}{4} = 7$

$log_2 x + \frac{1}{2} log_2 x + \frac{1}{4} log_2 x = 7$

$(\times 4)$

$4log_2 x + 2log_2 x + log_2 x = 28$

$7log_2\, x = 28$

$log_2\, x = \frac{28}{7}$

$log_2\, x = 4$

Converting to exponential form gives:

$x = 2^4$

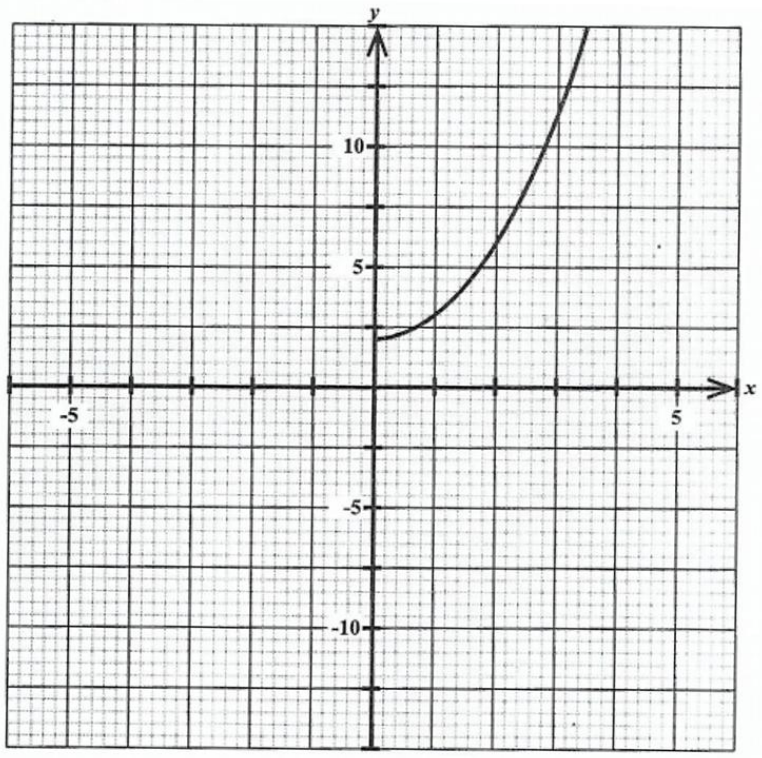
$x = 16$

$\therefore x = 16$

Total: 25 marks

June 2021 – Question 2 – Solution

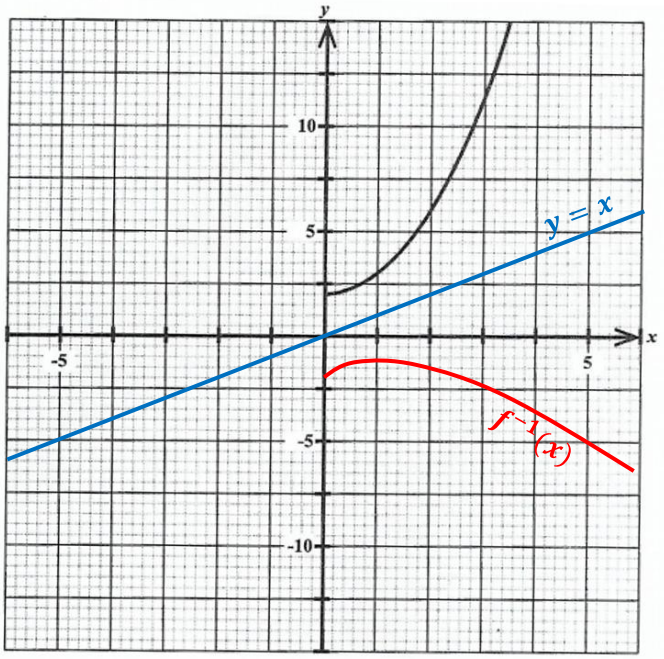
(a) The diagram below shows the graph of $f(x) = x^2 + 2$ for $x \geq 0$.



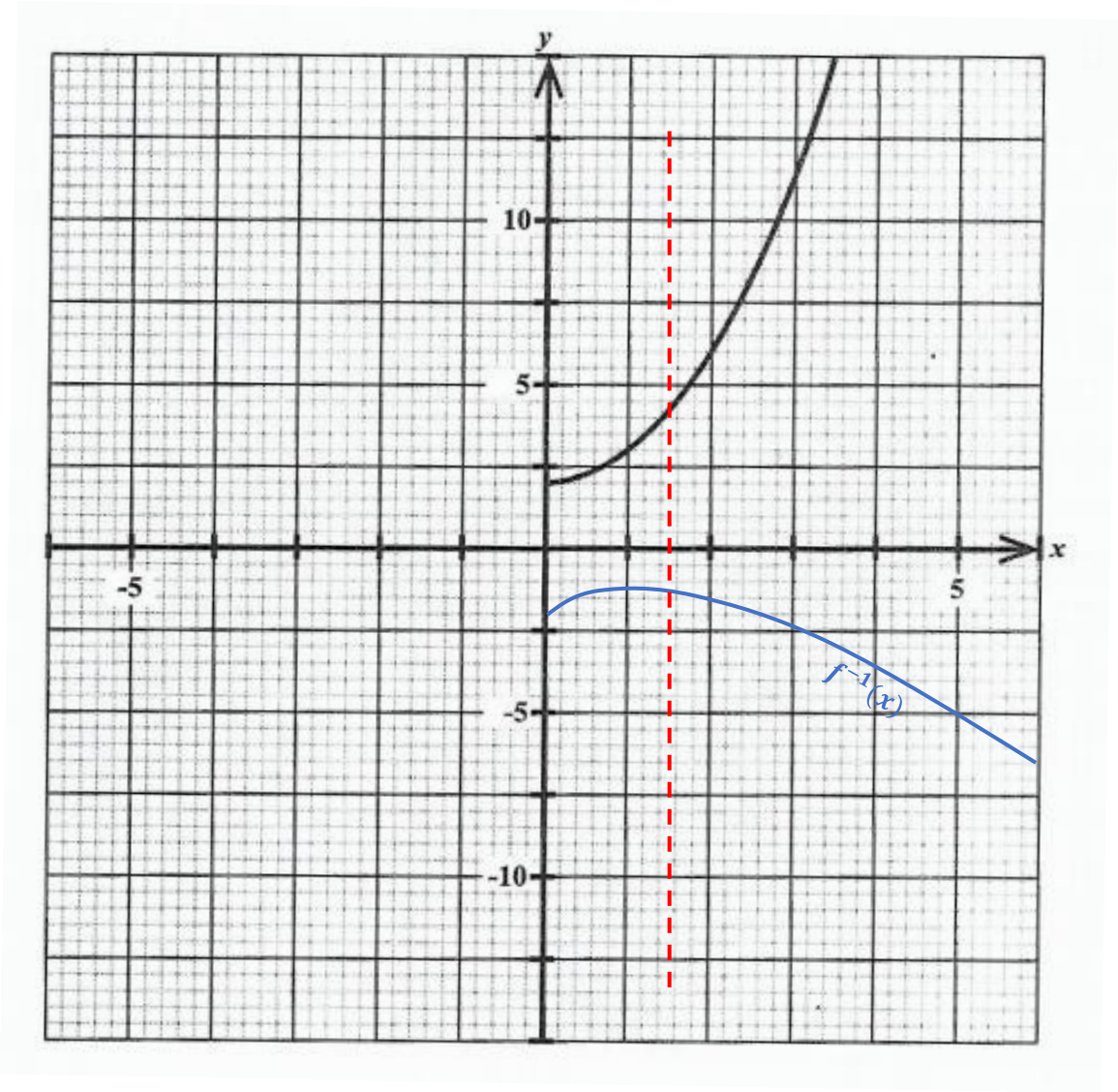
(i) On the graph,

(a) Sketch the inverse of f

The graph of the inverse of f is a reflection of the graph of f in the line $y = x$.



(b) Show that the inverse of f is a function. [4]

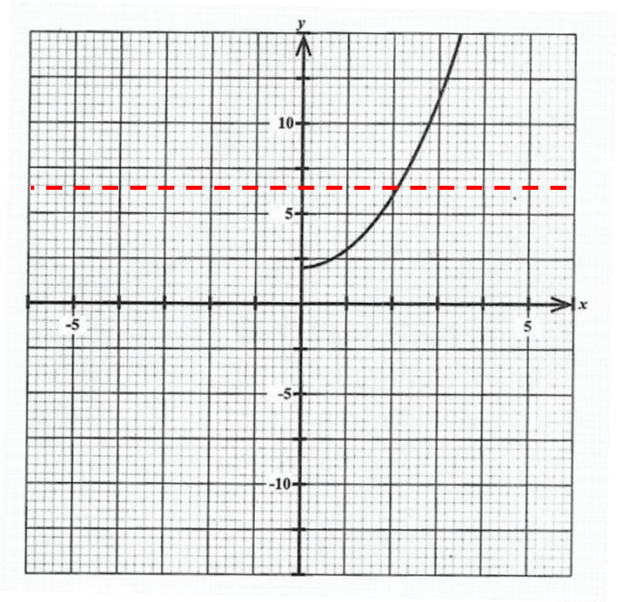


Any line drawn parallel to the y-axis cuts the graph at most once. In other words, for every x-value, there will be at most one y-value. This proves bijectivity.

Hence, the inverse of f is a function.

(ii) Prove that f is one to one. [4]

Graphical Proof:



Any line drawn parallel to the x-axis cuts the graph at most once. In other words, for every y-value, there will be at most one x-value.

Hence, f is a one-to-one function.

Proof using the definition:

$f(x) = x^2 + 2$ for $x \geq 0$.

Now,

$f(a) = a^2 + 2$

$f(b) = b^2 + 2$

Suppose that $f(a) = f(b)$.

Then, we have

$$a^2 + 2 = b^2 + 2$$

$$a^2 = b^2$$

$$a^2 - b^2 = 0$$

$$(a - b)(a + b) = 0$$

Either $a - b = 0$ or $a + b = 0$

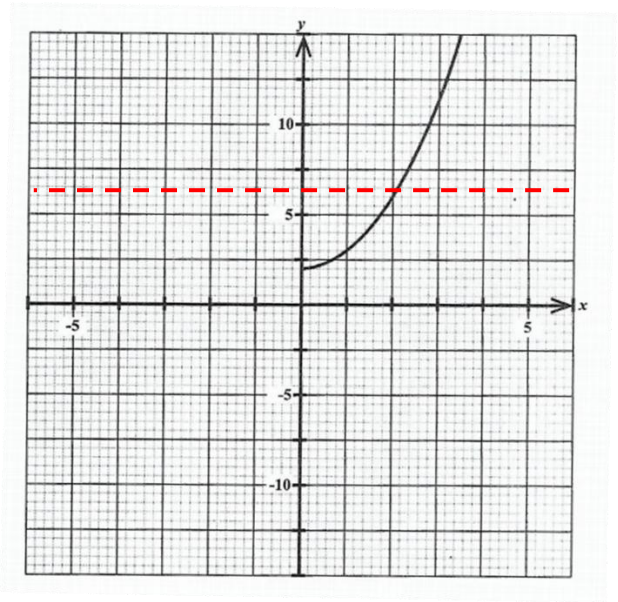
$a = b$ $a = -b$ which is invalid since $x \geq 0$

Since $f(a) = f(b)$ implies that $a = b$, then f is one-to-one.

Q.E.D.

(iii) Determine whether f is onto. [3]

Graphical Proof:



Any line drawn parallel to the x-axis cuts the graph at least once. In other words, for every y-value, there will be at least one x-value.

Hence, the graph shows that $f(x) = x^2 + 2$ for $x \geq 0$ is an onto function.

<u>Proof using the definition:</u>

$f(x) = x^2 + 2\,, x \geq 0.$

Let y be an element of the codomain.

$y = x^2 + 2$

Making x the subject of the formula gives:

$y - 2 = x^2$

$x = \sqrt{y - 2}$

From the graph, the codomain is $y \geq 2, y \in \mathbb{R}$.

Hence, for every y in the codomain, there is a corresponding x in the domain.

Therefore, $f(x)$ is an onto function.

(b) Solve the equation $|x^2 - 4| = 3x - 2$. [6]

$|x^2 - 4| = 3x - 2$

By definition:

$x^2 - 4 = 3x - 2$ or $-(x^2 - 4) = 3x - 2$

$x^2 - 3x - 2 = 0$ $-x^2 + 4 = 3x - 2$

$x^2 + 3x - 6 = 0$

<u>Case 1:</u> $x^2 - 3x - 2 = 0$

Using the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-2)}}{2(1)}$$

$$= \frac{3 \pm \sqrt{9+8}}{2}$$

Either $\quad x = \frac{3}{2} + \frac{\sqrt{17}}{2} \quad$ or $\quad x = \frac{3}{2} - \frac{\sqrt{17}}{2}$

<u>Case 2:</u> $x^2 + 3x - 6 = 0$

Using the quadratic formula,

$$x = \frac{-3 \pm \sqrt{(3)^2 - 4(1)(-6)}}{2(1)}$$

$$= \frac{-3 \pm \sqrt{9+24}}{2}$$

$$= \frac{-3 \pm \sqrt{33}}{2}$$

Either $\quad x = -\frac{3}{2} + \frac{\sqrt{33}}{2}$ or $\quad x = -\frac{3}{2} - \frac{\sqrt{33}}{2}$

Checking whether the values of x are valid.

$\therefore x = \frac{3}{2} + \frac{\sqrt{17}}{2}$ and $x = -\frac{3}{2} + \frac{\sqrt{33}}{2}$

(c) Given that α, β, and γ are the roots of the equation $2x^3 - x^2 + 3x - 1 = 0$,

determine the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. [8]

$2x^3 - x^2 + 3x - 1 = 0$

where $a = 2, b = -1, c = 3$ and $d = -1$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$= \frac{3}{2}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

For the new roots:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{3}{2} \div \frac{1}{2}$$

$$= \frac{3}{2} \times \frac{2}{1}$$

$$= 3$$

$$\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right)+\left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right)+\left(\frac{1}{\alpha}\right)\left(\frac{1}{\gamma}\right)=\frac{\gamma+\alpha+\beta}{\alpha\beta\gamma}$$

$$=\frac{1}{2}\div\frac{1}{2}$$

$$=1$$

$$\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right)=\frac{1}{\alpha\beta\gamma}$$

$$=1\div\frac{1}{2}$$

$$=1\times 2$$

$$=2$$

$\therefore$ The new equation is: $x^3-3x^2+x-2=0$

Total: 25 marks

June 2021 – Question 3 – Solution

(a) (i) Express $4\sin\theta + 3\cos\theta$ in the form $R\sin(\theta + \alpha)$. [5]

Let $4\sin\theta + 3\cos\theta = R\sin(\theta + \alpha)$

$$4\sin\theta + 3\cos\theta = R\sin\theta\cos\alpha + R\cos\theta\sin\alpha$$

Equating coefficients gives,

$R\cos\alpha = 4$ $\rightarrow$ Eq. 1

$R\sin\alpha = 3$ $\rightarrow$ Eq. 2

Eq. 2 ÷ Eq. 1 gives,

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{3}{4}$$

$$\tan\alpha = \frac{3}{4}$$

$$\alpha = 0.644^c$$

$$R = \sqrt{4^2 + 3^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25}$$

$$= 5$$

$$\therefore 4\sin\theta + 3\cos\theta = 5\sin(\theta + 0.644^c)$$

(ii) Hence, solve the equation $4\sin\theta + 3\cos\theta = 2$ for $0 \le \theta \le 2\pi$. [5]

$$4\sin\theta + 3\cos\theta = 2$$

$$5\sin(\theta + 0.644^c) = 2$$

$$\sin(\theta + 0.644^c) = \frac{2}{5}$$

$$\theta + 0.644^c = 2.730^c, 6.695^c$$

$$\theta = 2.09^c, 6.05^c$$

Note that:

$0 \le \theta \le 2\pi$

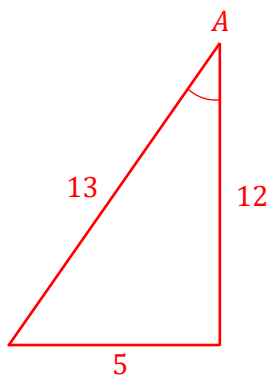
$0.644^c \le \theta + 0.644^c \le 2\pi + 0.644^c$

+	0.412

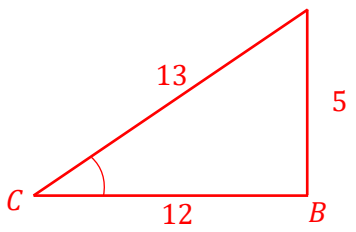
$\therefore \theta = 2.09^c, 6.05^c$ for $0 \le \theta \le 2\pi$

(b) Given that $\cos A = \frac{12}{13}$, $\sin C = \frac{5}{13}$, where A and C are acute angles, calculate the value of $\cos(A - C)$. [7]

We are given that $\cos A = \frac{12}{13}$ and $\sin C = \frac{5}{13}$.



From this triangle, we have $\cos A = \frac{12}{13}$ and $\sin A = \frac{5}{13}$.



From this triangle, we have $\sin C = \frac{5}{13}$ and $\cos C = \frac{12}{13}$.

Hence,

$$\cos(A - C) = \cos A \cos C + \sin A \sin C$$

$$= \left(\frac{12}{13}\right)\left(\frac{12}{13}\right) + \left(\frac{5}{13}\right)\left(\frac{5}{13}\right)$$

$$= \frac{144}{169} + \frac{25}{169}$$

$$= \frac{169}{169}$$

$$= 1$$

(c) Solve the equation $\cos\theta = \sin\left(\theta - \frac{\pi}{3}\right)$ for θ. [8]

$$\cos\theta = \sin\left(\theta - \frac{\pi}{3}\right)$$

By applying the addition formula on the right hand side, we get:

$$\cos\theta = \sin\theta\cos\frac{\pi}{3} - \cos\theta\sin\frac{\pi}{3}$$

$$\cos\theta = \sin\theta\left(\frac{1}{2}\right) - \cos\theta\left(\frac{\sqrt{3}}{2}\right)$$

$$2\cos\theta = \sin\theta - \sqrt{3}\cos\theta$$

$$2\cos\theta + \sqrt{3}\cos\theta = \sin\theta$$

$$\left(2 + \sqrt{3}\right)\cos\theta = \sin\theta$$

$$2 + \sqrt{3} = \frac{\sin\theta}{\cos\theta}$$

$$\tan\theta = 2 + \sqrt{3}$$

$$\theta = \tan^{-1}(2 + \sqrt{3})$$

$$\theta = \frac{5\pi}{12}$$

$\therefore$ The value of θ for which $\cos\theta = \sin\left(\theta - \frac{\pi}{3}\right)$ is $\theta = \frac{5\pi}{12}$.

Total: 25 marks

June 2021 – Question 4 – Solution

(a) A circle has equation $x^2 + y^2 - 4x + 10y - 8 = 0$.

(i) Determine the centre and radius of the circle. [4]

$$x^2 + y^2 - 4x + 10y - 8 = 0$$

$$x^2 - 4x + 4 + y^2 + 10y + 25 = 8 + 4 + 25$$

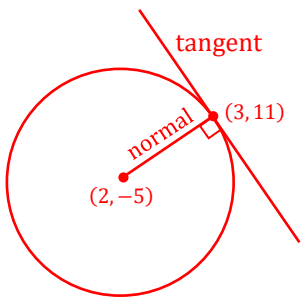
$$(x-2)^2 + (y+5)^2 = 37$$

which is of the form $(x-a)^2 + (y-b)^2 = r^2$

where the centre is $(2, -5)$ and the radius, $r = \sqrt{37}$ units.

(ii) Determine the equation of the tangent which touches the circle at (3,11). [4]

Consider the diagram below:



The normal to the circle has points $(2, -5)$ and $(3, 11)$.

Gradient of normal $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{11-(-5)}{3-2}$

$= \frac{16}{1}$

$= 16$

Since the product of the gradient of the normal and its tangent is -1, then the gradient of the tangent $= -\frac{1}{16}$.

Substituting $m = -\frac{1}{16}$ and point $(3, 11)$ into $y - y_1 = m(x - x_1)$, we get:

$$y - 11 = -\frac{1}{16}(x - 3)$$

$$y - 11 = -\frac{1}{16}x + \frac{3}{16}$$

$$y = -\frac{1}{16}x + \frac{179}{16}$$

$\therefore$ The equation of the tangent that touches the circle at $(3, 11)$ is:

$$16y = -x + 179$$

(b) Show that the curve whose parametric equations are $x = 3 + 3\sin\theta$ and $y = 3\cos\theta$ represents a circle. [5]

Rearranging the parametric equations, we have:

$x = 3 + 3\sin\theta$	and	$y = 3\cos\theta$
$x - 3 = 3\sin\theta$		$\frac{y}{3} = \cos\theta$
$\frac{x-3}{3} = \sin\theta$		$\left(\frac{y}{3}\right)^2 = \cos^2\theta$
$\left(\frac{x-3}{3}\right)^2 = \sin^2\theta$		

Now,

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{x-3}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$

$\frac{(x-3)^2}{9} + \frac{y^2}{9} = 1$

$\frac{(x-3)^2+y^2}{9} = 1$

$(x-3)^2 + y^2 = 9$

which is of the form $(x-a)^2 + (y-b)^2 = r^2$

where the centre of the circle is $(3, 0)$ and $r = 3$ units.

$\therefore$ The curve represents a circle.

(c) Two position vectors are represented by $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$.

Determine the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [8]

We are given that $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$.

<u>Recall:</u>

$\overrightarrow{OA} \cdot \overrightarrow{OB} = |\overrightarrow{OA}||\overrightarrow{OB}| \cos\theta$ where θ represents the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$.

Well,

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$$

$$= (1)(1) + (2)(-3) + (3)(2)$$

$$= 1 - 6 + 6$$

$$= 1$$

$$|\overrightarrow{OA}| = \sqrt{(1)^2 + (2)^2 + (3)^2}$$

$$= \sqrt{1 + 4 + 9}$$

$$= \sqrt{14}$$

$$|\overrightarrow{OB}| = \sqrt{(1)^2 + (-3)^2 + (2)^2}$$

$$= \sqrt{1 + 9 + 4}$$

$$= \sqrt{14}$$

Now,

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = |\overrightarrow{OA}||\overrightarrow{OB}| \cos\theta$$

$$1 = \sqrt{14}\sqrt{14} \cos\theta$$

$$\cos\theta = \frac{1}{\sqrt{14}\sqrt{14}}$$

$$\theta = \cos^{-1}\left(\frac{1}{14}\right)$$

$$\theta = 85.9° \text{ (to 1 decimal place)}$$

(d) Determine the vector equation of a plane which passes through the point $(2, 5, 3)$ and is perpendicular to the vector $4\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$. [4]

The vector equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 4 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 4 \\ -1 \end{pmatrix}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 4 \\ 4 \\ -1 \end{pmatrix} = (2)(4) + (5)(4) + (3)(-1)$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 4 \\ 4 \\ -1 \end{pmatrix} = 8 + 20 - 3$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 4 \\ 4 \\ -1 \end{pmatrix} = 25 \qquad \text{or} \qquad \boldsymbol{r} \cdot \left(4\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}\right) = 25$$

Total: 25 marks

June 2021 – Question 5 – Solution

(a) Use the substitution $u = (x^3 + 4)$ to determine $\int 3x^2(x^3 + 4)^4\, dx$. [6]

Let $u = x^3 + 4$

$$\frac{du}{dx} = 3x^2$$

$$dx = \frac{du}{3x^2}$$

Now,

$$\int 3x^2(x^3 + 4)^4\, dx = \int 3x^2 u^4 \frac{du}{3x^2}$$

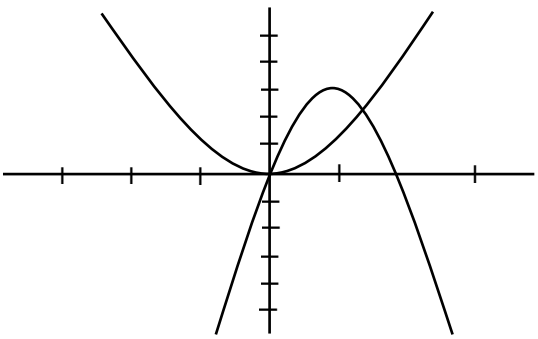
$$= \int u^4\, du$$

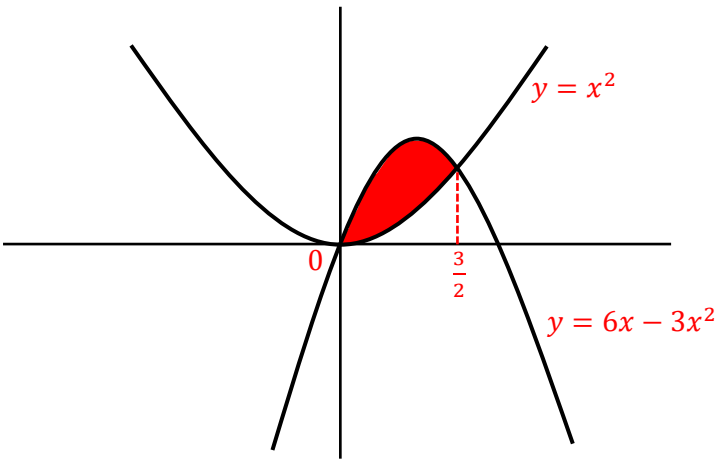
$$= \frac{u^5}{5} + c$$

$$= \frac{(x^3+4)^5}{5} + c$$

(b) Calculate the area of the region enclosed by the parabolas $y = x^2$ and $y = 6x - 3x^2$.

[A sketch of the curves is shown below.] [9]





$y = x^2$ → Equation 1

$y = 6x - 3x^2$ → Equation 2

Equating both equations gives:

$$x^2 = 6x - 3x^2$$

$$3x^2 + x^2 - 6x = 0$$

$$4x^2 - 6x = 0$$

$$2x^2 - 3x = 0$$

$$x(2x - 3) = 0$$

Either $x = 0$ or $2x - 3 = 0$

$$x = \frac{3}{2}$$

Now,

Area of shaded region = Area under maximum curve − Area under minimum curve

The limits are 0 and $\frac{3}{2}$.

Area under maximum curve $= \int_a^b y\,dx$

$$= \int_0^{\frac{3}{2}} (6x - 3x^2)\,dx$$

$$= \left[\frac{6x^2}{2} - \frac{3x^3}{3}\right]_0^{\frac{3}{2}}$$

$$= [3x^2 - x^3]_0^{\frac{3}{2}}$$

$$= \left[3\left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^3\right] - [3(0)^2 - (0)^3]$$

$$= 3\left(\frac{9}{4}\right) - \frac{27}{8} - 0$$

$$= \frac{27}{4} - \frac{27}{8} - 0$$

$$= \frac{27}{8}\ \text{units}^2$$

Area under minimum curve $= \int_a^b y\,dx$

$$= \int_0^{\frac{3}{2}} x^2\,dx$$

$$= \left[\frac{x^3}{3}\right]_0^{\frac{3}{2}}$$

$$= \frac{1}{3}[x^3]_0^{\frac{3}{2}}$$

$$= \frac{1}{3}\left[\left(\frac{3}{2}\right)^3 - (0)^3\right]$$

$$= \frac{1}{3}\left(\frac{27}{8} - 0\right)$$

$$= \frac{9}{8}\ \text{units}^2$$

Hence,

Area of shaded region $= \frac{27}{8} - \frac{9}{8}$

$$= \frac{9}{4}\,\text{units}^2$$

(c) The equation of a curve is given as $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$.

(i) Show that $f(x)$ has stationary points at $x = -1, x = 0$ and $x = 2$. [6]

$f(x) = 3x^4 - 4x^3 - 12x^2 + 1$

$f'(x) = 12x^3 - 12x^2 - 24x$

At stationary points, $f'(x) = 0$.

When $f'(x) = 0$,

$12x^3 - 12x^2 - 24x = 0$

$x^3 - x^2 - 2x = 0$

$x(x^2 - x - 2) = 0$

$x(x+1)(x-2) = 0$

Either $x = 0$ or $x + 1 = 0$ or $x - 2 = 0$

$x = -1$ $x = 2$

$\therefore f(x)$ has stationary points at $x = -1, x = 0$ and $x = 2$.

Q.E.D.

(ii) Determine the nature of these stationary points. [4]

$f(x) = 3x^4 - 4x^3 - 12x^2 + 1$

$f'(x) = 12x^3 - 12x^2 - 24x$

$f''(x) = 36x^2 - 24x - 24$

When $x = -1$,

$f''(-1) = 36(-1)^2 - 24(-1) - 24$

$= 36 + 24 - 24$

$= 36 \quad (> 0)$

Since $f''(-1) > 0$, then there is a minimum point at $x = -1$.

When $x = 0$,

$f''(0) = 36(0)^2 - 24(0) - 24$

$= 0 - 0 - 24$

$= -24 \quad (< 0)$

Since $f''(0) < 0$, then there is a maximum point at $x = 0$.

When $x = 2$,

$f''(2) = 36(2)^2 - 24(2) - 24$

$= 144 - 48 - 24$

$= 72 \quad (> 0)$

Since $f''(2) > 0$, then there is a minimum point at $x = 2$.

Total: 25 marks

June 2021 – Question 6 – Solution

(a) A function f is defined as $f(x) = \begin{cases} 2x+3 & ; x < 1 \\ 2 & ; x = 1 \\ \frac{x^2-1}{x-1} & ; x > 1 \end{cases}$

(i) Determine whether or not the $\lim_{x \to 1} f(x)$ exists. [5]

For the $\lim_{x \to 1} f(x)$ to exist, $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x)$.

Now,

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (2x+3)$$

$$= 2(1) + 3$$

$$= 2 + 3$$

$$= 5$$

And,

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \frac{x^2-1}{x-1}$$

$$= \lim_{x \to 1^+} \frac{(x-1)(x+1)}{(x-1)}$$

$$= \lim_{x \to 1^+} (x+1)$$

$$= 1 + 1$$

$$= 2$$

Since $\lim_{x \to 1^-} f(x) = 5$ and $\lim_{x \to 1^+} f(x) = 2$, then $\lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$.

$\therefore$ The $\lim_{x \to 1} f(x)$ does not exist.

(ii) Determine whether f is continuous at $x = 1$. [2]

Since $\lim_{x \to 1} f(x)$ does not exist, then f is not continuous at $x = 1$.

(b) The parametric equation of a curve is given by $x = 5\cos\theta$ and $y = 2\sin\theta$.

Determine $\frac{dy}{dx}$ in terms of θ. [4]

$x = 5\cos\theta$ $\qquad$ $y = 2\sin\theta$

$\frac{dx}{d\theta} = -5\sin\theta$ $\qquad$ $\frac{dy}{d\theta} = 2\cos\theta$

Using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= 2\cos\theta \times \frac{1}{-5\sin\theta}$$

$$= -\frac{2}{5}\frac{\cos\theta}{\sin\theta}$$

$$= -\frac{2}{5}\cot\theta$$

(c) Given that $\int_1^4 f(x)\,dx = \frac{10}{3}$, $\int_1^2 f(x)\,dx = 2$ and $g(x) = +\sqrt{x}$, calculate

$\int_2^4 [f(x) + g(x)]\,dx$ [4]

$$\int_2^4 [f(x) + g(x)]\,dx = \int_2^4 f(x)\,dx + \int_2^4 g(x)\,dx$$

$$= \int_1^4 f(x)\,dx - \int_1^2 f(x)\,dx + \int_2^4 \sqrt{x}\,dx$$

$$= \frac{10}{3} - 2 + \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_2^4$$

$$= \frac{4}{3} + \frac{2}{3}\left[(4)^{\frac{3}{2}} - (2)^{\frac{3}{2}}\right]$$

$$= \frac{4}{3} + \frac{2}{3}(8 - 2\sqrt{2})$$

$$= \frac{4}{3} + \frac{16}{3} - \frac{4}{3}\sqrt{2}$$

$$= \frac{20}{3} - \frac{4}{3}\sqrt{2}$$

$$= \frac{4}{3}(5 - \sqrt{2})$$

(d) (i) Solve the differential equation $\frac{dy}{dx} = \frac{\sin x}{\sin y}$ given that when $x = 0$, $y = \frac{\pi}{2}$. [6]

$$\frac{dy}{dx} = \frac{\sin x}{\sin y}$$

$$\sin y\,dy = \sin x\,dx$$

$$\int \sin y\,dy = \int \sin x\,dx$$

$$-\cos y = -\cos x + c$$

When $x = 0$ and $y = \frac{\pi}{2}$,

$$-\cos\left(\frac{\pi}{2}\right) = -\cos(0) + c$$

$$0 = -1 + c$$

$$c = 1$$

Hence,

$$-\cos y = -\cos x + 1$$

$$\cos y = \cos x - 1$$

$\therefore$ The differential equation is: $\cos y = \cos x - 1$

(ii) Determine the equation of the curve that passes through $(1, 5)$ and for which

$y = \int 6x^2\, dx$. [4]

$$y = \int 6x^2\, dx$$

$$= \frac{6x^3}{3} + c$$

$$= 2x^3 + c$$

When $x = 1$ and $y = 5$,

$5 = 2(1)^3 + c$

$5 = 2 + c$

$c = 5 - 2$

$c = 3$

$\therefore$ The equation of the curve is: $y = 2x^3 + 3$

Total: 25 marks